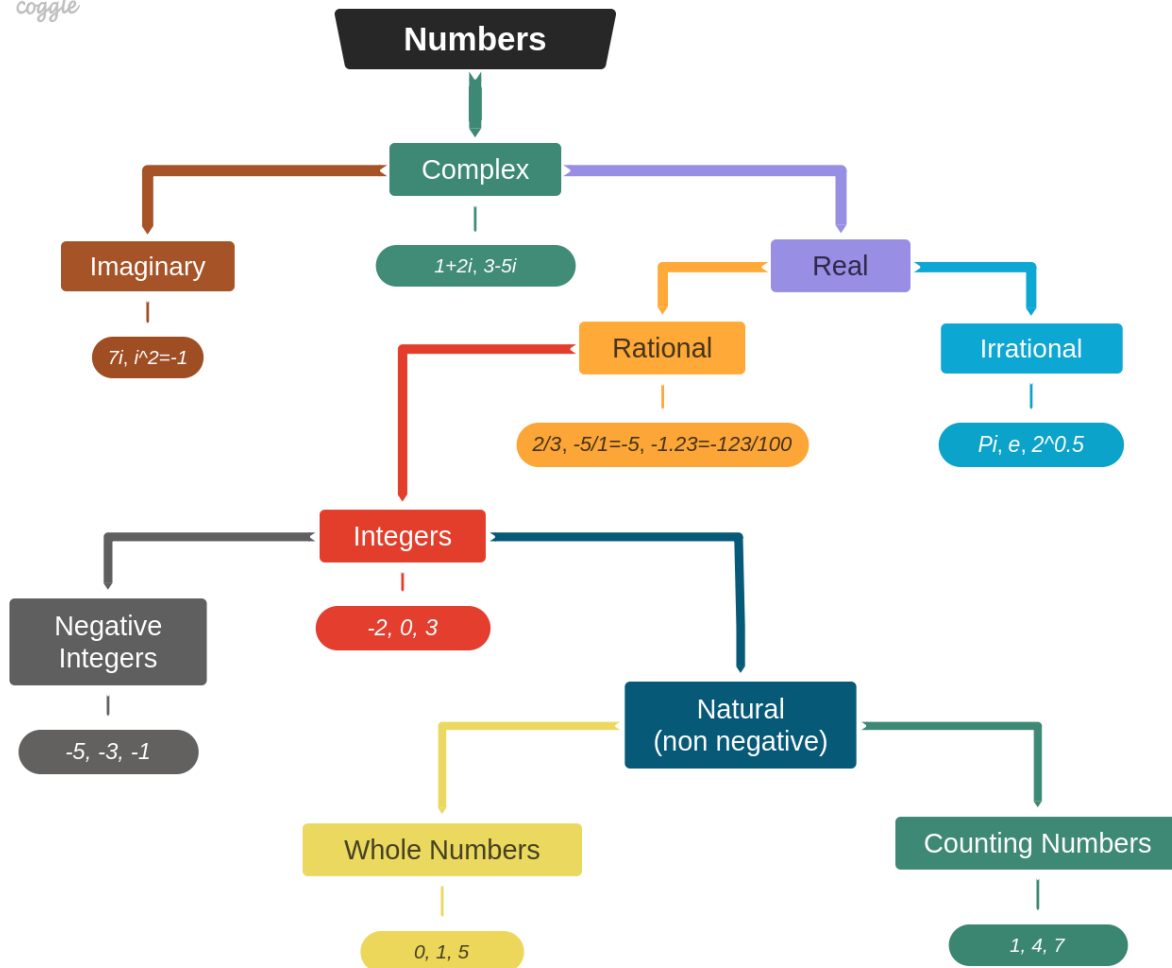


Number System

coggle



1. Complex Numbers ($\mathbb{C}$)

- General form: $a + bi$, where
 - a = real part
 - b = imaginary part
 - $i = \sqrt{-1}$
- Example: $3 + 2i$, $-5 + i$, 7 (when $b=0 \rightarrow$ purely real)

→Types:

- **Real Numbers** ($b = 0$)
 - **Purely Imaginary Numbers** ($a = 0$, e.g. $5i$, $-7i$)
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2. Real Numbers ($\mathbb{R}$)

- Numbers that can be represented on the number line.
- Includes **Rational + Irrational numbers**.

→Types of Real Numbers:

(a) Rational Numbers ($\mathbb{Q}$)

- Form: p/q , where $p, q \in \text{Integers}$, $q \neq 0$
- Decimal form: **terminating** or **repeating**.
- Examples: $1/2$, $-3/4$, 5 , $0.333\dots$
- **Subtypes:**
 1. **Integers** ($\mathbb{Z}$) $\rightarrow \dots -3, -2, -1, 0, 1, 2, 3 \dots$
 - **Whole Numbers** ($\mathbb{W}$) $\rightarrow 0, 1, 2, 3, \dots$
 - **Natural Numbers** ($\mathbb{N}$) $\rightarrow 1, 2, 3, \dots$
 2. **Fractions** $\rightarrow$ proper, improper, mixed fractions.
 3. **Recurring Decimals** $\rightarrow 0.333\dots, 2.142142\dots$

(b) Irrational Numbers

- Cannot be expressed as p/q .
- Decimal expansion: **non-terminating, non-repeating**.
- Examples: $\sqrt{2}$, $\sqrt{3}$, π , e .

3. Imaginary Numbers

- Numbers involving $\sqrt{-1} = i$.
 - Examples: i , $2i$, $-7i$.
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4. Special Sets of Numbers

- **Prime Numbers** → greater than 1, divisible only by 1 and itself. (2, 3, 5, 7, 11 ...)
 - **Composite Numbers** → more than two factors. (4, 6, 9 ...)
 - **Even Numbers** → divisible by 2. (... , -4, -2, 0, 2, 4 ...)
 - **Odd Numbers** → not divisible by 2. (... , -3, -1, 1, 3, 5 ...)
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Divisibility Rules

Quick tests without actual division:

- By 2 → Last digit even (0, 2, 4, 6, 8)
- By 3 → Sum of digits divisible by 3
- By 4 → Last 2 digits divisible by 4
- By 5 → Last digit 0 or 5
- By 6 → Divisible by both 2 and 3
- By 8 → Last 3 digits divisible by 8
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- By 9 $\rightarrow$ Sum of digits divisible by 9
- By 11 $\rightarrow$ (Sum of odd-position digits – Sum of even-position digits) divisible by 11

Power Cycle (Last Digit of Powers)

When powers increase, last digits repeat in cycles.

□ Method:

- Find last digit of base.
- Identify cycle length.
- Divide power by cycle length to find position.

Common Cycles:

- Last digit 2 $\rightarrow$ 2, 4, 8, 6 (cycle length 4)
- Last digit 3 $\rightarrow$ 3, 9, 7, 1 (length 4)
- Last digit 4 $\rightarrow$ 4, 6 (length 2)
- Last digit 7 $\rightarrow$ 7, 9, 3, 1 (length 4)
- Last digit 8 $\rightarrow$ 8, 4, 2, 6 (length 4)
- Last digit 9 $\rightarrow$ 9, 1 (length 2)

Examples

1. Find last digit of 7^{35}

- Cycle for 7 $\rightarrow$ (7,9,3,1) length 4
- $35 \div 4 \rightarrow$ remainder 3 $\rightarrow$ 3rd digit = 3

- Find last digit of 12^{58} Last digit = 2 $\rightarrow$ cycle (2,4,8,6)
- $58 \div 4 \rightarrow$ remainder 2 $\rightarrow$ 2nd digit = **4**

Remainder Cycle:

Remainder cycle is entirely different from unit digit cycle. But the same process which is used in unit digit cycle can be used to find out remainder cycle.

i.e., we need to find the **pattern for** $R\left(\frac{x^n}{y}\right)$,

E.g.: What will be the remainder when 4^{92} is divided by 7?

$$R\left(\frac{4^1}{7}\right) = 4$$

$$R\left(\frac{4^2}{7}\right) = 2$$

$$R\left(\frac{4^3}{7}\right) = 1$$

$$R\left(\frac{4^4}{7}\right) = 4$$

So, we can say that the remainder series of powers of 4 is 3
i.e., the cycle repeats itself after three values. (4, 2 and 1)

$R(\frac{92}{3}) = 2$. So, substitute 2 instead of

$$\therefore R(\frac{4^{92}}{7}) = R(\frac{4^2}{7}) = 2$$

Factors:

Factors of a number are the numbers which will exactly divide the given number without leaving any remainders

Factors of **12** = {1, 2, 3, 4, 6, 12}

Properties of Factors:

1. Every number has **1** and the number itself as factors.
(e.g., $7 \rightarrow$ factors are 1 and 7).
2. Factors are always $\leq$ **the number**.
3. Every number has a **finite number** of factors.
4. If a number has exactly two factors (1 and itself), it is a **prime number**.

Multiples

- **Definition:**

A **multiple** of a number is obtained by multiplying it with natural numbers (1, 2, 3, ...).

- **Example:**

Multiples of **4** = {4, 8, 12, 16, 20, ...}

Properties of Multiples:

1. The smallest multiple of a number is the number itself.
(e.g., smallest multiple of 7 is 7).
2. Every number has **infinite multiples**.
3. Zero (0) is a multiple of every number.
4. Multiples are always $\geq$ **the number**.

1. Highest Common Factor (HCF)

- **Definition:**

The largest number that divides exactly (without remainder) two or more given numbers.

Also called **GCD (Greatest Common Divisor)**.

- **Methods to find HCF:**

1. **Prime Factorization Method:**

Write prime factors of each number → Take common factors with the smallest power → Multiply them.

Example:

Find HCF of **18, 24**.

- $18 = 2 \times 3^2$

- $24 = 2^3 \times 3$

Common factors = $2^1 \times 3^1 = 6$

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2. Lowest Common Multiple (LCM)

- **Definition:**

The smallest number that is divisible by two or more given numbers.

- **Methods to find LCM:**

1. **Prime Factorization Method:**

Write prime factors of each number → Take all factors with the **highest powers** → Multiply them.

Example:

Find LCM of **18, 24**.

- $18 = 2 \times 3^2$

- $24 = 2^3 \times 3$

LCM = $2^3 \times 3^2 = 72$

NOTE:

Type	Approach
Find the greatest number that will exactly divide a, b and c	HCF (a, b, c)
Find the greatest number that will divide a, b and c leaving remainders of x, y and z respectively.	HCF (a - x, b - y, c - z)
Find the greatest number which when it divides a, b and c will leave the same remainder in each case.	HCF (a - b, b - c, c - a)
Find the least number which is exactly divisible by a, b and c.	LCM (a, b, c)
Find the least number which when divided by a, b and c leaves the same remainder 'r' in each case.	LCM (a, b, c) + r

Practice Problems

Q1. Is 735 divisible by 5?

Sol: A number is divisible by 5 if its last digit is 0 or 5. Here last digit = 5 $\rightarrow$ divisible.

Q2. Check divisibility of 472 by 11.

Sol: Rule: Difference of (sum of odd place digits) – (sum of even place digits). = $(4+2) - (7) = -1 \rightarrow$ Not divisible.

Q3. Find smallest number to be added to 5743 to make it divisible by 9.

Sol: Sum of digits = 19 $\rightarrow$ remainder when divided by 9 = 1. So add = $9-1=8$.

Q4. Which is divisible by 8: 624, 917, 1352?

Sol: Last 3 digits check. 624 and 1352 divisible, 917 not.

Q5. Check if 37845 is divisible by 15.

Sol: Rule: Divisible by both 3 & 5. Sum=27 divisible by 3, last digit=5 divisible by 5 $\rightarrow$ divisible.

Q6. Find remainder when 3271 is divided by 9.

Sol: Sum=13, remainder=4.

Q7. Find smallest 4-digit number divisible by 21.

Sol: $1000 \div 21 = \text{remainder } 13$, so number = 1008.

Q8. Is 10101 divisible by 99?

Sol: Must be divisible by 9 and 11. Fails divisibility by 9 $\rightarrow$ not divisible.

Q9. Find last digit of 7^{35} .

Sol: Cycle 7,9,3,1 length=4. $35 \bmod 4 = 3 \rightarrow 3^{\text{rd}}=3$.

Q10. Last digit of 9^{48} ?

Sol: Cycle 9,1 length=2. $48 \bmod 2 = 0 \rightarrow \text{last digit}=1$.

Q12. Last digit of 2^{121} .

Sol: Cycle 2,4,8,6 length=4. $121 \bmod 4=1 \rightarrow \text{digit}=2$.

Q13. Last digit of 15^{93} .

Sol: Always ends with 5.

Q14. Last digit of 8^{67} .

Sol: Cycle 8,4,2,6 length=4. $67 \bmod 4=3 \rightarrow \text{digit}=2$.

Q15. Last digit of 11^{2025} .

Sol: Always ends with 1.

Q17. Find remainder when $2^{25} \div 7$.

Sol: Cycle 2,4,1 length=3. $25 \bmod 3=1 \rightarrow \text{remainder}=2$.

Q18. Remainder of $3^{45} \div 5$.

Sol: Cycle 3,4,2,1 length=4. $45 \bmod 4=1 \rightarrow \text{remainder}=3$.

Q19. Remainder of $7^{103} \div 12$.

Sol: Cycle 7,1 length=2. Odd power $\rightarrow \text{remainder}=7$.

Q20. Find remainder when $2^{100} \div 13$.

Sol: Cycle length=12. $100 \bmod 12=4 \rightarrow \text{remainder}=8$.

Q21. Find number of factors of 120.

Sol: $120=2^3 \times 3 \times 5$. Factors $=(3+1)(1+1)(1+1)=16$.

Q22. Find sum of factors of 36.

Sol: $36=2^2 \times 3^2$. Sum $=(1+2+4)(1+3+9)=91$.

Q23. Find greatest 3-digit multiple of 27.

Sol: $999 \div 27=37 \rightarrow \text{divisible}$. Ans=999.

Q24. How many multiples of 9 between 100 and 1000?

Sol: First=108, last=999. Count $=(999-108)/9+1=100$.

Q25. Find total even factors of 360.

Sol: $360=2^3 \times 3^2 \times 5$. Even factors $=(3)(3)(2)=18$.

Q26. Find HCF of 252 and 198.

Sol: $252=2^2 \times 3^2 \times 7$, $198=2 \times 3^2 \times 11$. HCF=18.

Q27. Smallest number divisible by 12 and 18.

Sol: $\text{LCM}=36$.

Q28. Find number of factors of 5040.

Sol: $5040=2^4 \times 3^2 \times 5 \times 7$. Factors $= (5)(3)(2)(2)=60$.

Q29. Find HCF of 48 and 64.

Sol: $48=2^4 \times 3$, $64=2^6$. HCF $=16$.

Q30. Find LCM of 18 and 24.

Sol: $18=2 \times 3^2$, $24=2^3 \times 3$. LCM $=72$.

Q31. Two numbers are 72 and 120. Find $\text{HCF} \times \text{LCM}$.

Sol: $=\text{Product}=8640$.

Q32. Find HCF of 27, 63, 81.

Sol: $27=3^3$, $63=3^2 \times 7$, $81=3^4$. HCF $=9$.

Q33. LCM of 25, 30, 40?

Sol: $25=5^2$, $30=2 \times 3 \times 5$, $40=2^3 \times 5$. LCM $=600$.

Q34. Smallest number divisible by 12,18,24 with remainder 3?

Sol: LCM $=72$. $\rightarrow 72k+3 \rightarrow 75$.

Q35. Three bells ring 12,15,20 mins. Together again?

Sol: LCM $=60$. $\rightarrow 1 \text{ hr} \rightarrow 8 \text{ AM}$.

Q36. Greatest 4-digit divisible by 15,20,25?

Sol: LCM $=300$. Largest $=9900$.